

Cybersecurity

Introduction to Windows and CMD

Lesson 7.1



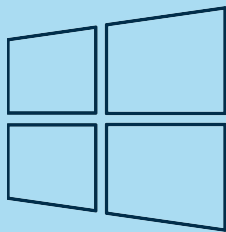
Learning Objectives

By the end of today's class, you will be able to:

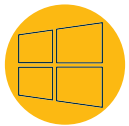
- 1 Leverage the Windows Command Prompt (CMD) to navigate and manage directories and files.
- 2 Explain how Windows stores files.
- 3 Create, manage, and view user information using the command-line tool **net**.
- 4 Manage password policies using **gpedit**.
- 5 Optionally, schedule tasks using Task Scheduler.



Welcome

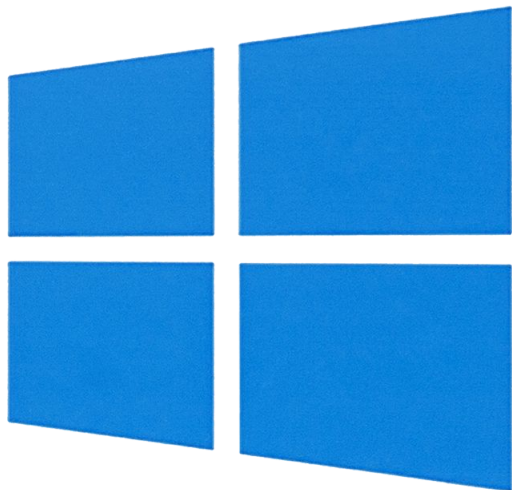


Introduction to **Windows**



Welcome to Windows

Windows is the leader for desktop operating systems.



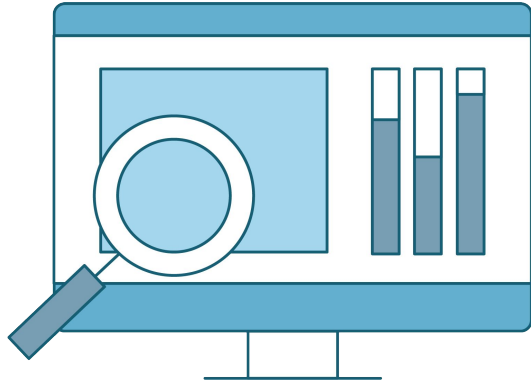
The popularity of Windows machines make them the **most common target**.



Windows knowledge is essential in many roles:

SOC Analyst

SOC analysts must monitor and detect suspicious activity on Windows devices.



System Administrator

Most system administrators work with one or more Microsoft products and/or services.



Windows knowledge is essential in many roles:

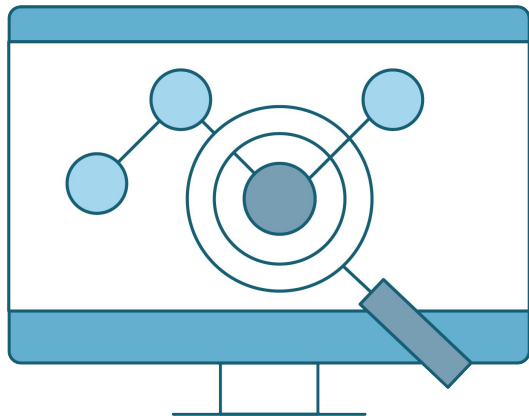
Penetration Tester

Penetration testers need to know how to exploit Windows and Microsoft-related products.



Endpoint Forensics

Forensics investigators have to have a detailed understanding of how Windows works.



Windows System Administrator

Today's activities will cover common system administration tasks:



Audit processes with Task Manager.



Use the command line to gather information and create files.



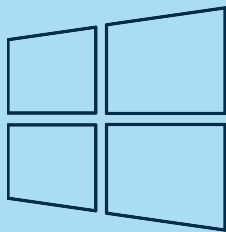
Enforce password policies.



Manage users.

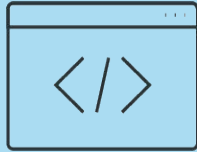


Automate tasks.



Launching Your Windows Lab

Before we get started, let's log into,
and launch our **Lab environment**.

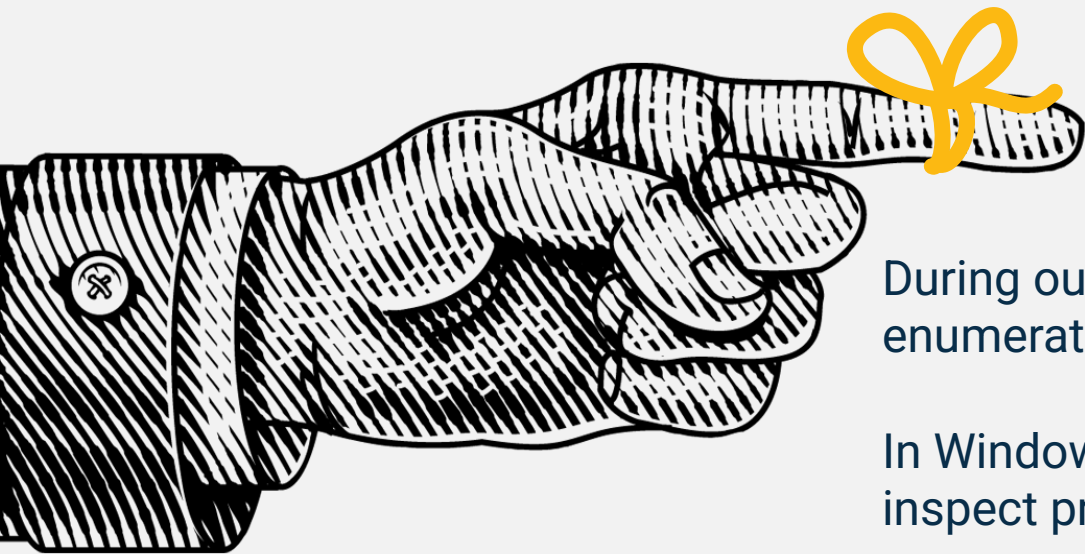


Introduction to **Task Manager**



Did you notice the **excessive number of processes** that started up when you logged into the Windows VM?

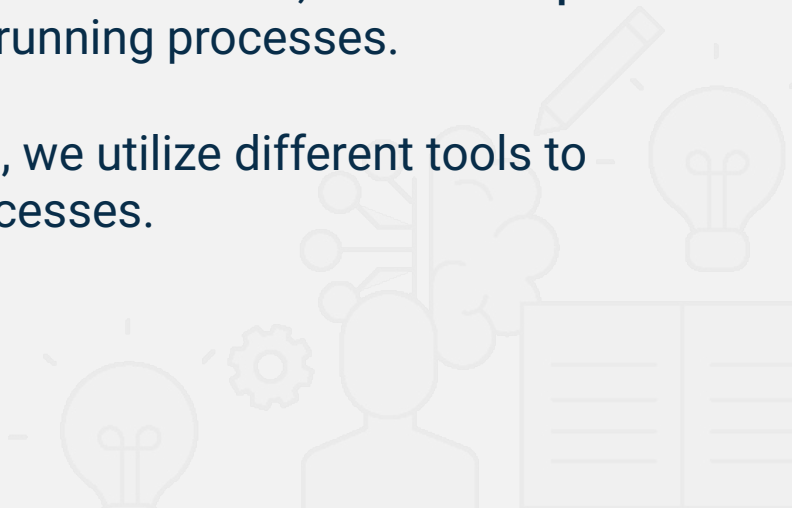
This is what a Windows workstation can be like if not maintained by an organization's system administrator.



Remember,

During our Linux modules, we used **top** to enumerate running processes.

In Windows, we utilize different tools to inspect processes.

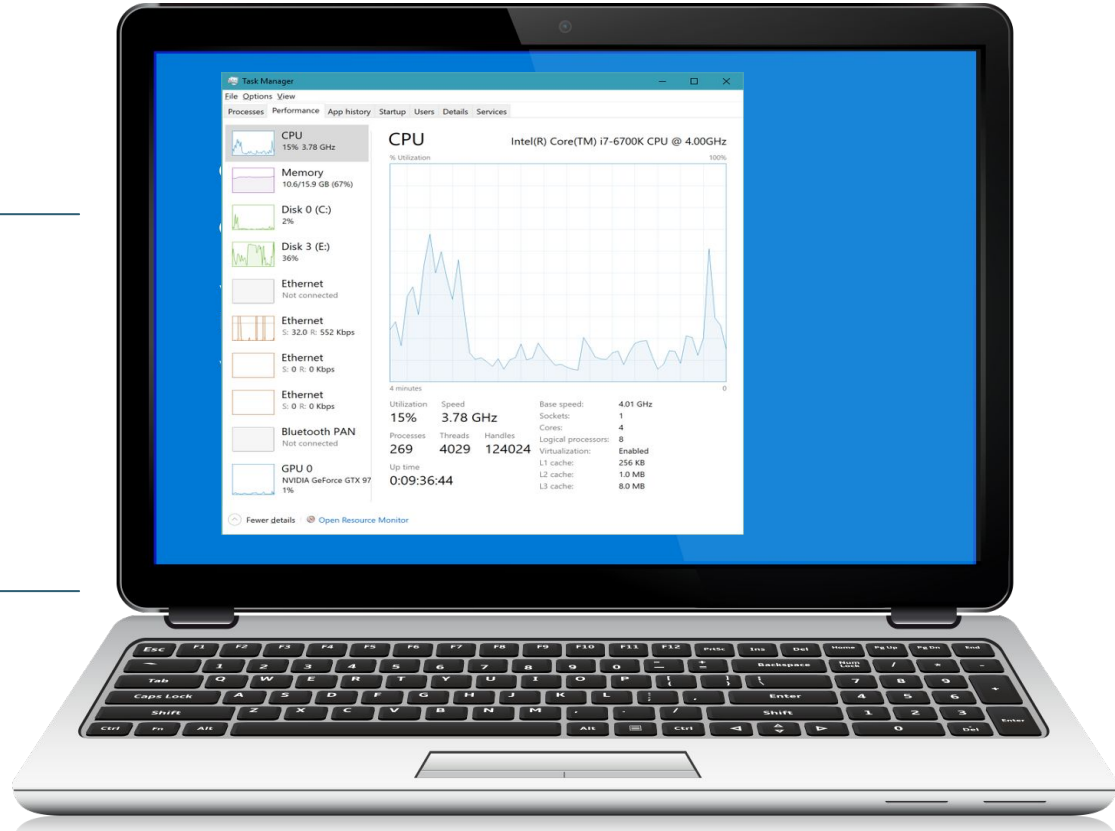


Task Manager

Windows tool that allows:

Auditing and managing processes to identify errant or malicious actions

Inspection of PIDs and startup processes

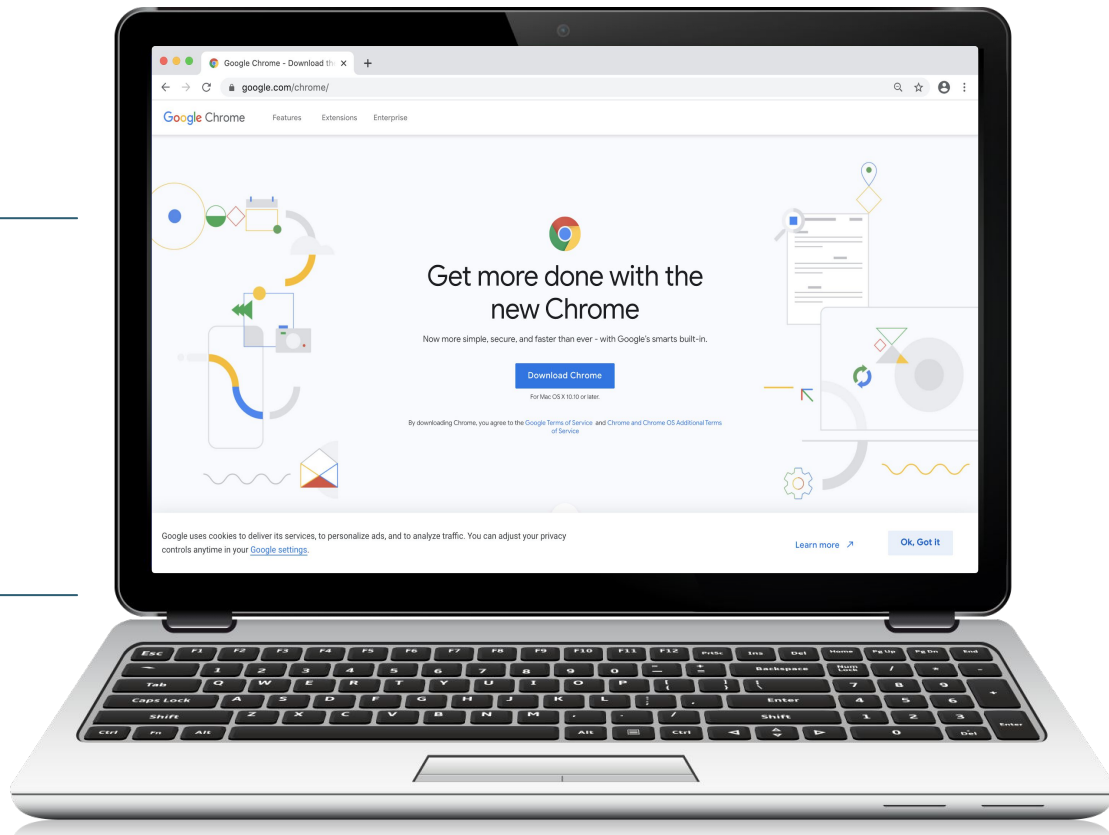


Task Manager

Practical uses for processes

Google Chrome, which is well-known for high memory usage

RDP applications leaving systems vulnerable



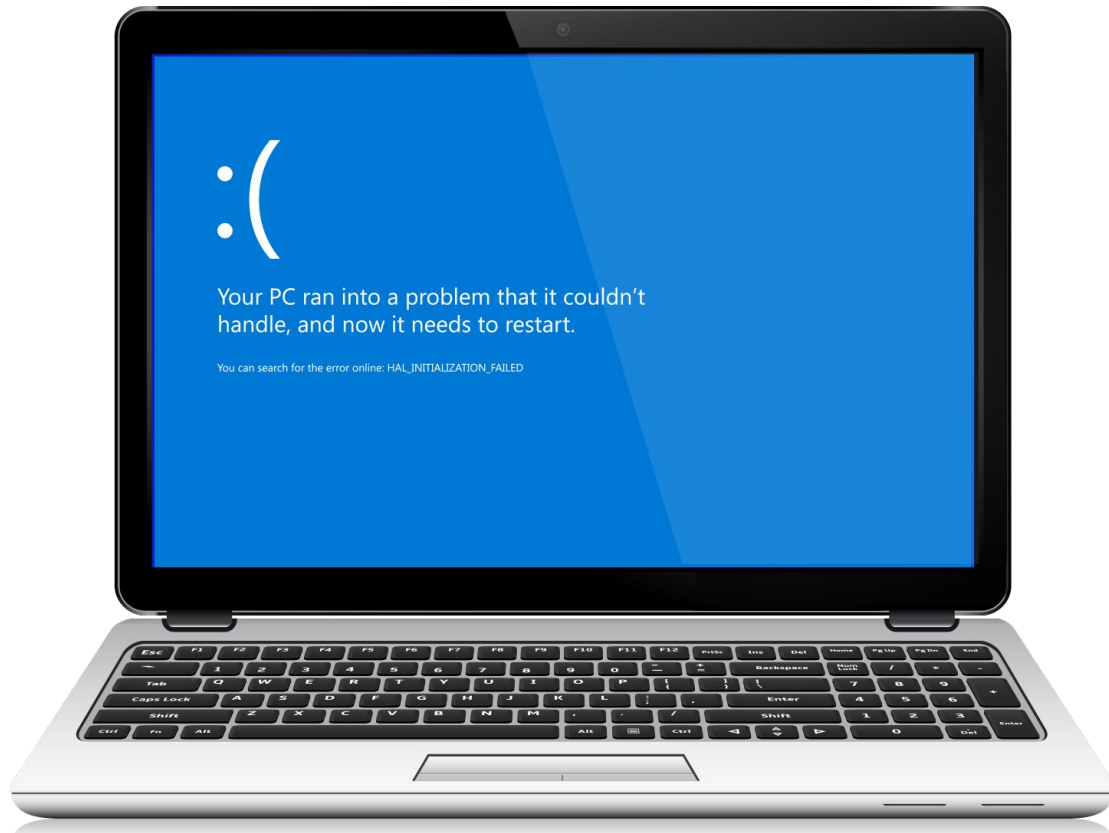
Task Manager

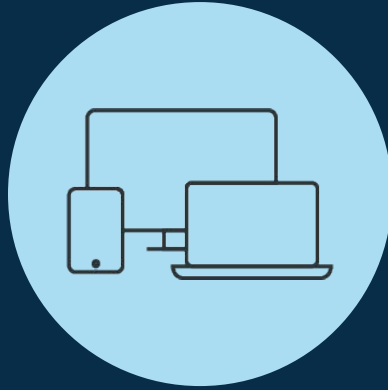
Blue Screen of Death

Some processes can cause:

- Memory leaks
- System instability
- System crashes

Blue Screen of Death



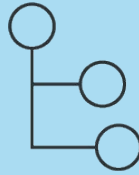


Instructor Demonstration

Task Manager Overview

Managing Startup Applications

-  Slow boot times
-  Excessive resource usage
-  Network bandwidth allocation
-  Applications may require special permissions



Windows **File** Structure



Windows File Structure

Similar to **Linux**, **Windows** uses a folder system on a drive with subdirectories

Storage devices are represented by drive letters.

Primary Drive often holds the OS, typically c:\.

```
C:\
├─ PerfLogs\
├─ Program Files\
├─ Program Files (x86)\
├─ ProgramData\ [hidden folder]
├─ Users\
│   └─ [username]\
│       ├── Desktop\
│       └─ Documents\
├─ Windows\
│   ├── System32\
│   │   ├── Config\
│   │   └─ Drivers\
│   │       └─ etc\
│   │           ├── hosts
│   │           └─ networks
│   └─ Spool
└─ Temp
```



Windows File Structure

Directories and Special Folders

Primary Drive



- C:\
- \“root” drive

Special folders



- Specific purpose folders
- Documents, Pictures, etc.

```
C:\
├─ PerfLogs\
├─ Program Files\
├─ Program Files (x86)\
├─ ProgramData\ [hidden folder]
├─ Users\
│   └─ [username]\
│       ├── Desktop\
│       └─ Documents\
└─ Windows\
    ├── System32\
    │   ├── Config\
    │   └─ Drivers\
    │       └─ etc\
    │           ├── hosts
    │           └─ networks
    ├── Spool
    └─ Temp
```



Windows File Structure

Attributes, Extensions, and NTFS

File Attributes

- Read-only
- Hidden
- System
- Applications



File Extensions

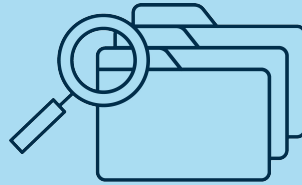
Identify the type of files and associate the applications



File System

New Technology File System (NTFS)



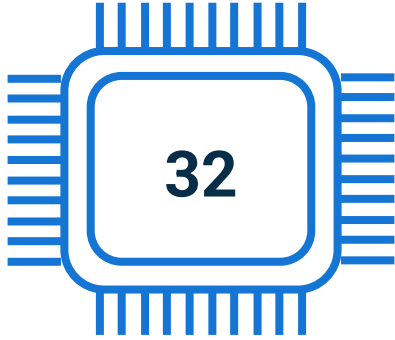


Windows **OS** Folders



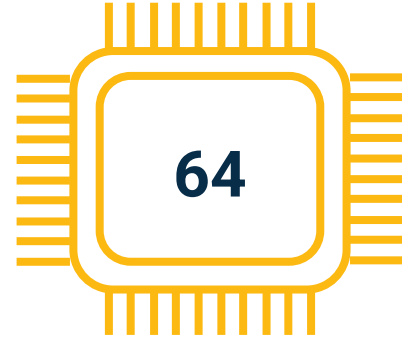
Before we did into the OS folders, we need to know the differences in **processor architecture.**

32-bit vs 64-bit Processors



32-Bit (x86) Processors

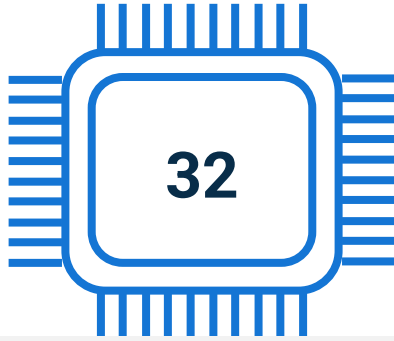
- Memory up to 4GB
- 64-bit Applications won't work
- System cache 1GB
- x86



64-Bit (x64) Processors

- Memory up to 16TB
- Runs 64-bit **and** 32-bit applications
- System cache 1TB
- x64

32-bit Operating Systems



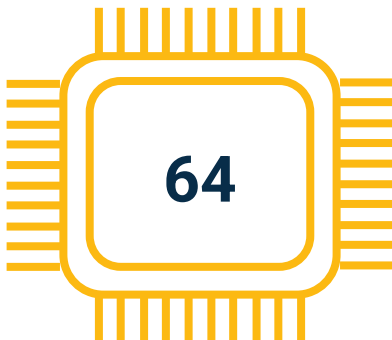
System Folder

`C:\Windows\System32`
Contains 32-bit system binaries

Program Folder

`C:\Program Files`
Contains 32-bit programs

64-bit Operating Systems



System Folder

C:\Windows\System32
Contains 64-bit system binaries

C:\Windows\SysWOW64
Contains 32-bit system binaries

Program Folder

C:\Program Files
Contains 64-bit programs

C:\Program Files (x86)
Contains 32-bit programs

System32 Folder

System32 is a critical Windows directory

System Files

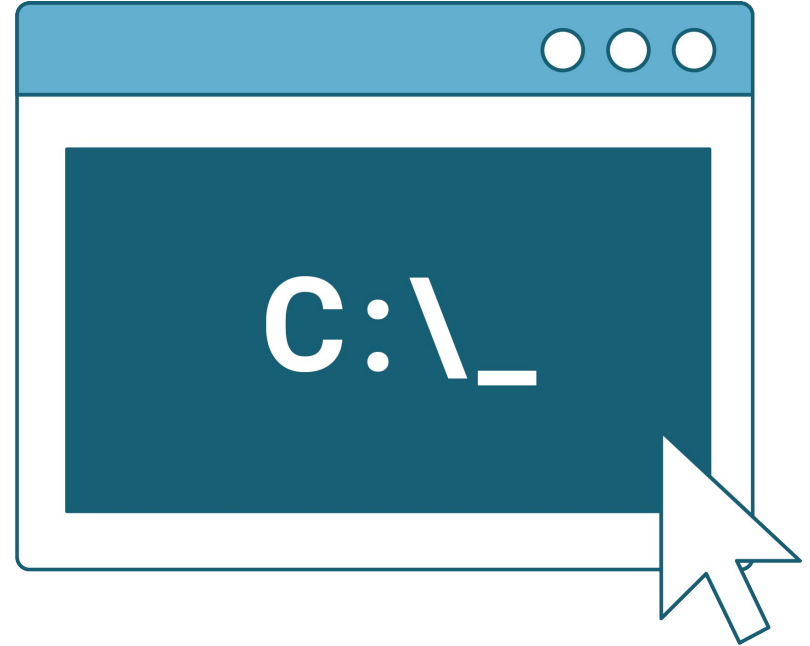
- Dynamic link libraries (DLLs)
- Executables (EXE)
- Device drivers

System Utilities

- `cmd.exe`
- `taskmgr.exe`
- `regedit.exe`

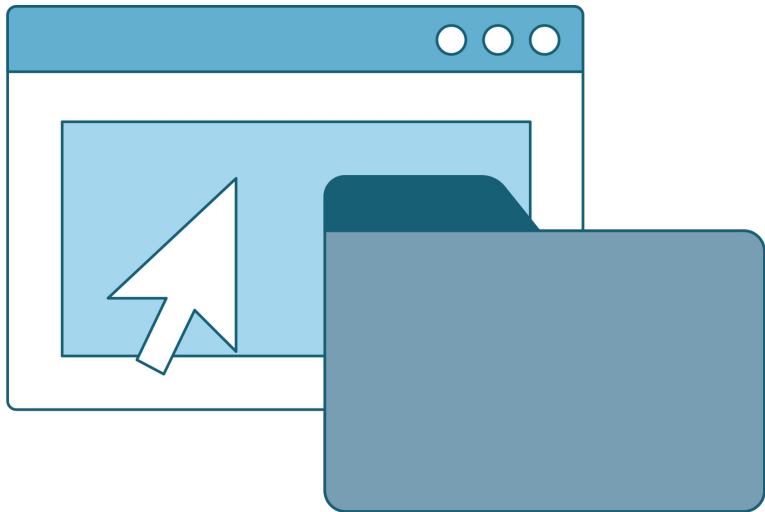
64-bit System Files

- 64-bit location
- SysWOW64 stores 32-bit



Program Files Folder

C:\Program Files and C:\Program Files (x86)



Application Installations

- Default directory

Program Data

- Data files
- Configuration files
- Documentation

User Account Control (UAC)

- Protects modification or deletion

Hidden Folders

Folder and directories in disguise

Visibility

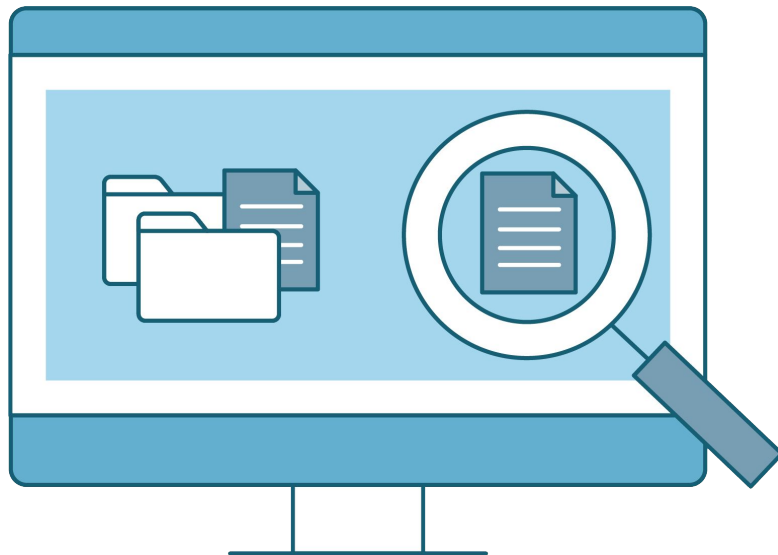
- Accidental deletion or modification
- AppData, ProgramData

Configuration Files

- Application settings
- Temporary files

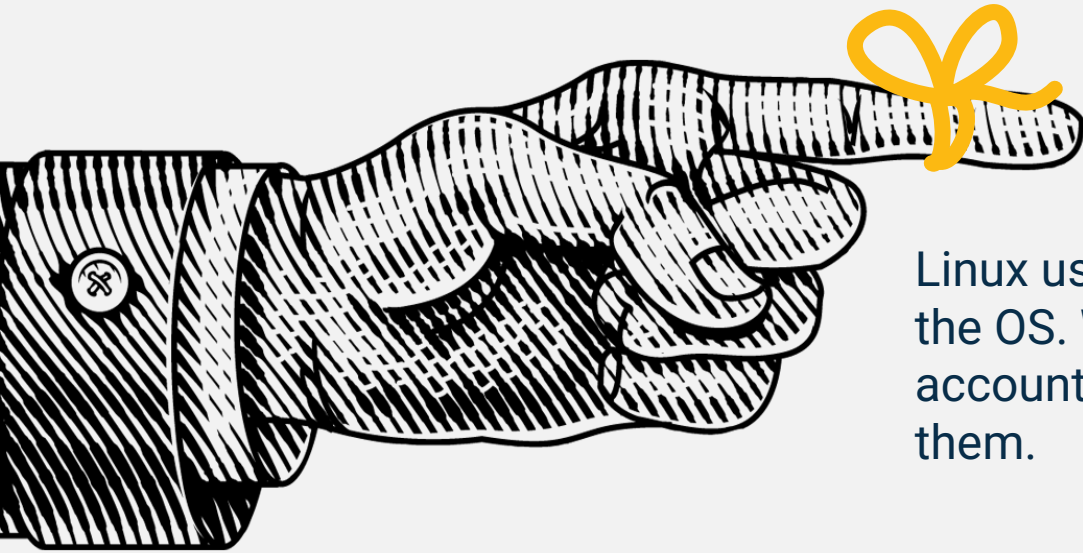
Viewing Hidden Folders

- Not visible by default
- “Show hidden files, folders, and drives”





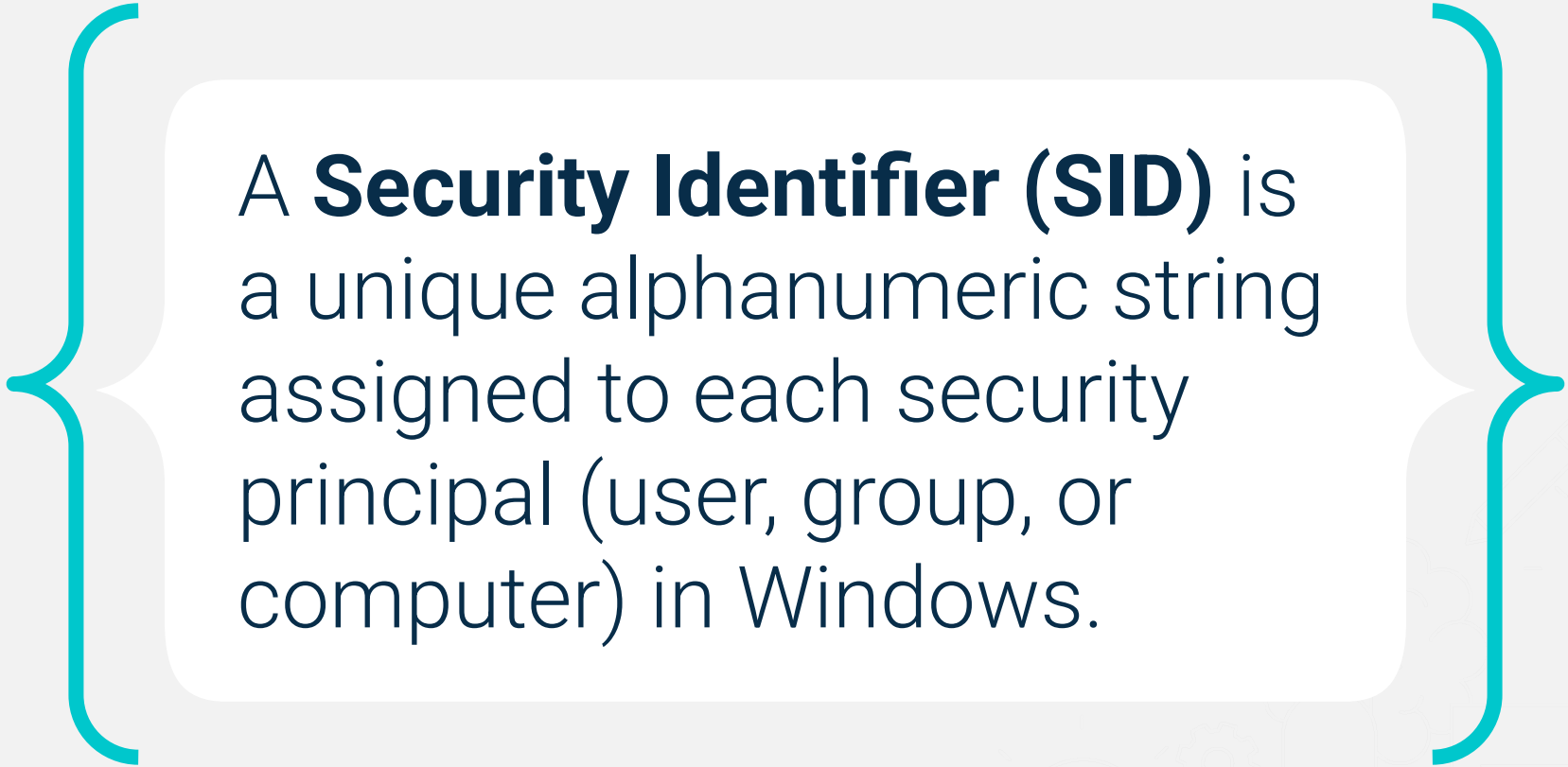
Windows **Users**



Remember,

Linux uses different account types within the OS. **Windows** also allows for different accounts and assigns **various privileges** to them.





A **Security Identifier (SID)** is a unique alphanumeric string assigned to each security principal (user, group, or computer) in Windows.



Windows Users

Security Identifier (SID)

Revision Number:

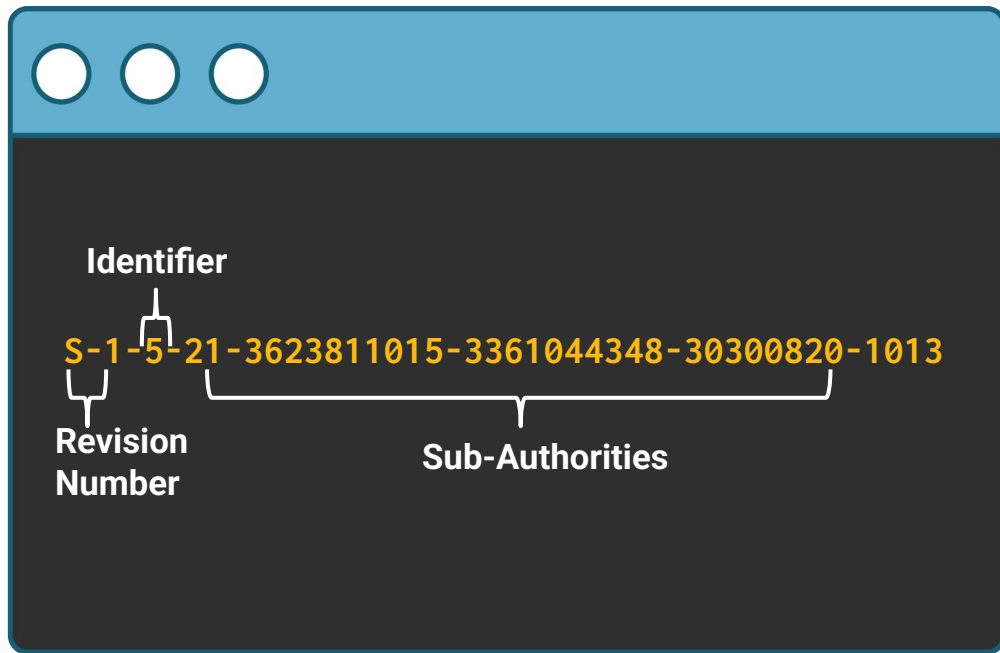
Revision level of SID, typically “S-1”

Identifier:

Authority issued to SID, in Windows always “5”

Sub-Authorities:

Variable number identifies SID within a domain or local system



Windows Users

Additional SID properties in networked environments:

-  Used extensively in access control lists (ACLs) to define permissions
-  Identify domain user accounts
-  SIDs determine user authorization level and access rights
-  Identify members of security groups



Windows Users

Well-known SIDS

01

Everyone: S-1-1-0 represents all users and groups, including anonymous users.

02

Authenticated Users: S-1-5-11 represents all users who have authenticated to the local or a trusted domain.

03

Local System: S-1-5-18 represents the built-in SYSTEM account with full system privileges.

04

Administrators: S-1-5-32-544 represents the built-in Administrators group.



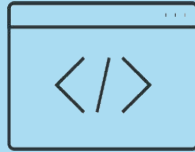
Windows Users

SYSTEM

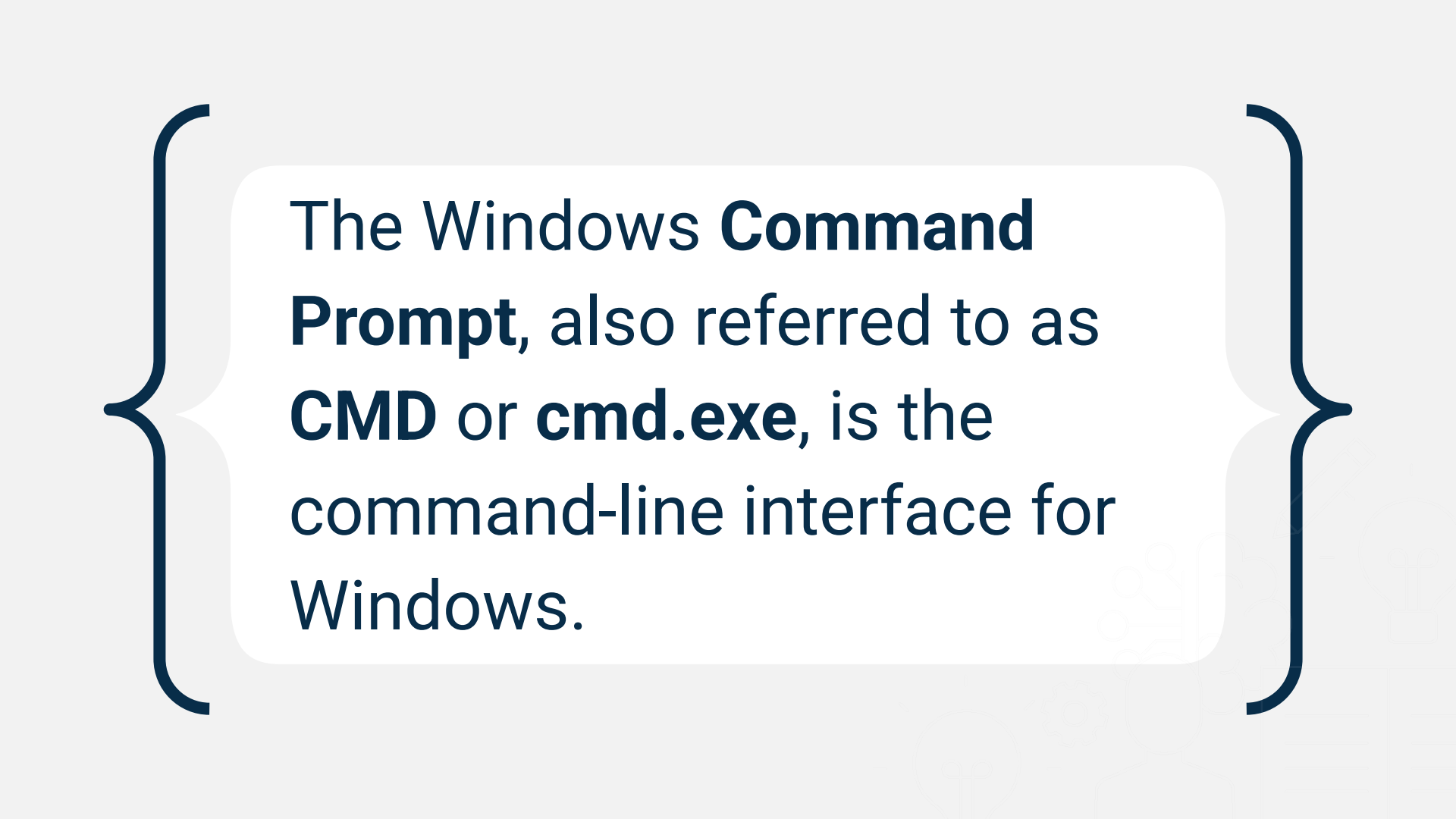
Similar to Linux's **root**, Windows uses the **SYSTEM** account.

- Used by OS itself
- Highest level of privileges
- Essential for core services and processes
- Runs critical services, drivers, and processes for:
 - Managing hardware
 - Kernel interactions
 - Executing operations





Introduction to **Command Prompt**



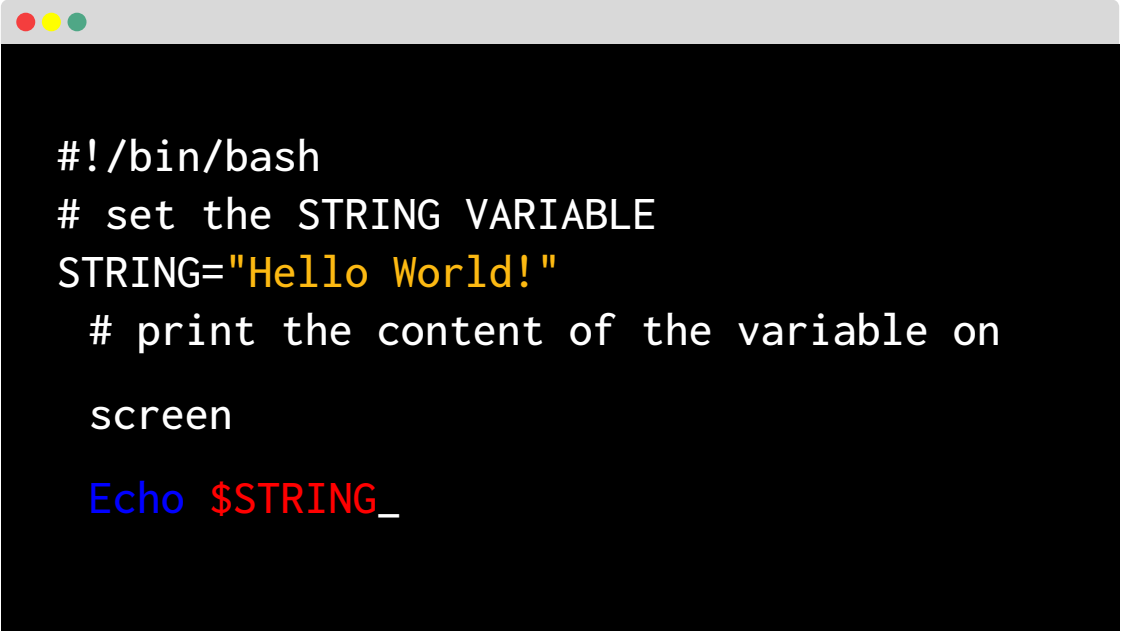
The Windows **Command Prompt**, also referred to as **CMD** or **cmd.exe**, is the command-line interface for Windows.



Remember environment
variables from the **Bash**
programming unit?

CMD and Environment Variables

In Windows, they work the same way—they're preset by the system and usable in the command line and scripts.



```
#!/bin/bash
# set the STRING VARIABLE
STRING="Hello World!"
# print the content of the variable on
screen

Echo $STRING_
```


Environment variables (envvars)

are special values that contain information about the system, such as the user's home directory or the system's program files directory.

Common ENV Variables

Envvars can be used for the following:

- Shortening long directory paths
- Grabbing the current time
- Finding the location of system files

Environment Variables	Default Value
%CD%	Current directory
%DATE%	Current date
%OS%	Windows
%ProgramFiles%	C:\Program Files
%ProgramFiles(x86)%	C:\Program Files (x86)
%TIME%	Current time
%USERPROFILE%	C:\Users\{username}
%SYSTEMDRIVE%	C:\
%SYSTEMROOT%	C:\Windows

Common ENV Variables

Linux variables are designated with a \$, while Windows ENV variables are enclosed with % signs.

For example, to navigate to the 64-bit **Program Files** folder, we run:

```
cd %ProgramFiles%
```

We can combine ENV variables with regular directory names:

```
cd %USERPROFILE%\Desktop
```

This would send us to the desktop of the current user.

Common ENV Variables

We can combine environment variables with regular directory names:

```
cd %USERPROFILE%\Desktop
```

%USERPROFILE% is a variable assigned to the value of the current user's home directory.



This is the same as \$HOME in Linux.

Command Prompt (CMD)

Windows Command Prompt (CMD or `cmd.exe`), is the command-line interface for Windows, comparable to a Unix shell, such as Bash for Linux.



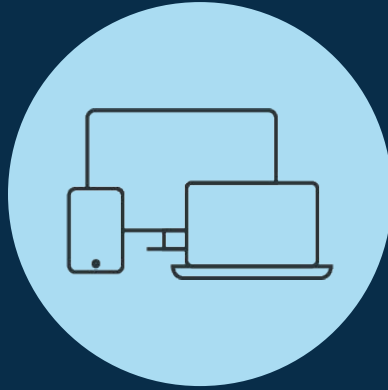
NOTE

Command prompts are not case-sensitive with files and directories.

`cd "Program files"` is the same as `cd "PROGRAM FILES"`.

Use quotes around the name of a file or directory that contains spaces.

CMD Command	Action	Linux Counterpart
<code>cd</code> or <code>chdir</code>	Change directory	<code>cd</code>
<code>dir</code>	List contents of directory	<code>ls</code>
<code>md</code> or <code>mkdir</code>	Create directory	
<code>copy</code>	Copy file	<code>cp</code>
<code>move</code>	Move (cut and paste) files	<code>mv</code>
<code>del</code> or <code>erase</code>	Delete files and directories	
<code>rd</code> or <code>rmdir</code>	Remove a directory if empty	
<code>find</code>	Search a file for specified string	
<code>exit</code>	Close CMD	
<code>type</code>	Show contents of specified file	<code>cat</code>



Instructor Demonstration

CMD Parameters and To-Do List



Activity:

Intro to Task Manager and CMD

In this activity, you will play the role of a junior systems administrator at Good Corp, Inc.

- Your CIO has requested that you start a baseline report about a Windows machine. In an upcoming lesson, this Windows workstation will be joined to an Active Directory Domain Controller.
- Your CIO has requested that you do the following on the Windows workstation:
 - Use Task Manager to clean up running and startup processes.
 - Use **CMD** to create a simple text file in the current user's Desktop directory called **report.txt**. You will later append Windows system information to it.

Suggested Time:
15 minutes



Time's up!
Let's review

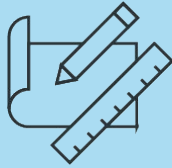


Questions?





Break
15 mins



Users, Groups, and Password **Policies**



Next, we'll use the
command-line tool **net**
to manage user
accounts, groups, and
password policies.

Using net

We'll be using the following `net` utilities:

`net user`

For adding, removing, and managing users.

`net localgroup`

For adding, removing, and managing local groups.

`net accounts`

For viewing password and logon requirements for users to enforce password security policies.



Using net

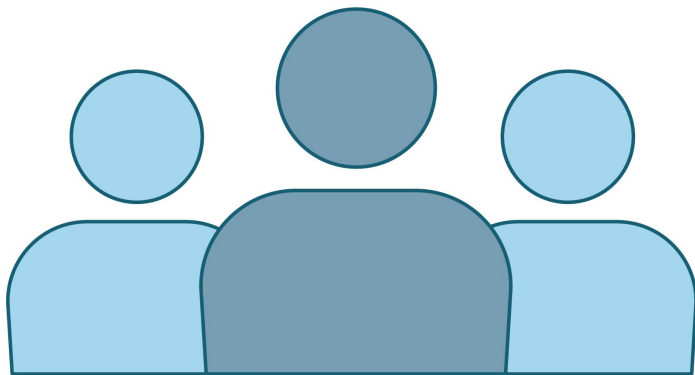
`net` can be used to set password policies

- The minimum number of characters required for a password
- The amount of time before passwords expire
- The minimum number of days before a password can be changed
- The number of times a password must be unique before it can be reused again





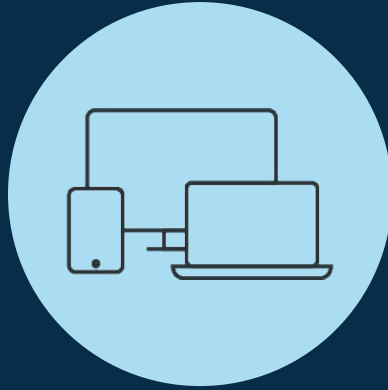
Demonstration Scenario



Your CIO was curious about the groups and password policies on the Windows workstation. You are to retrieve more information from this Windows workstation using the **net** command-line utility.

Use **net** to:

- Enumerate users to see **net** output.
- Enumerate **azadmins** group and password policies.
- Enumerate local groups.
- Enumerate Windows workstation current password policies.



Instructor Demonstration

net



Activity:

Users, Groups, and Password Policies

Your CIO is curious about what groups and password policies are on the Windows workstation.

- You are tasked with retrieving more information from this Windows workstation using the `net` command-line utility.

Suggested Time:
10 minutes

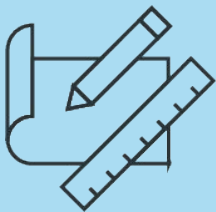


Time's up!
Let's review

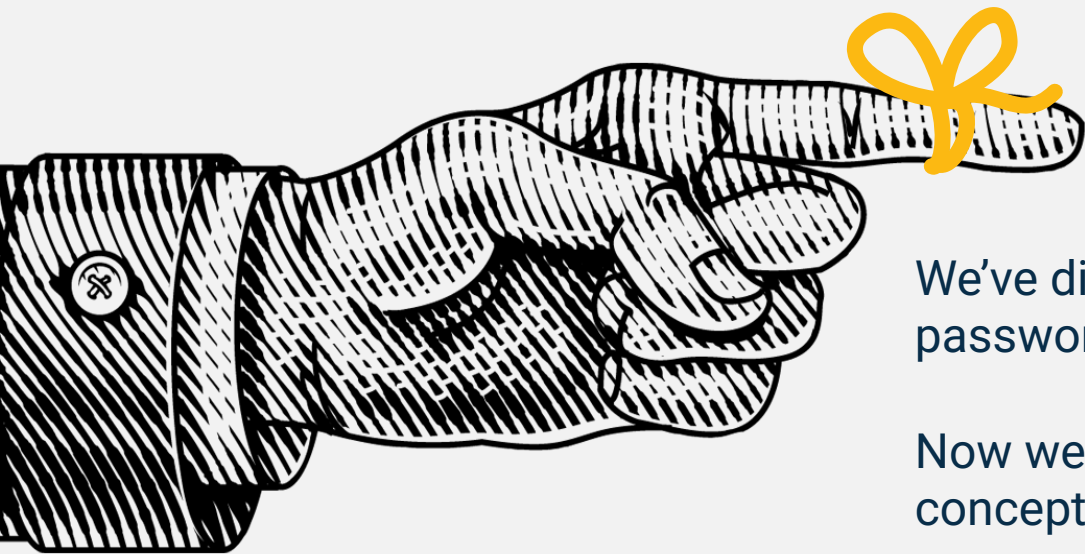


Questions?





Creating Users and Setting Password Policies



Remember,

We've discussed the importance of password policies in earlier Linux units.

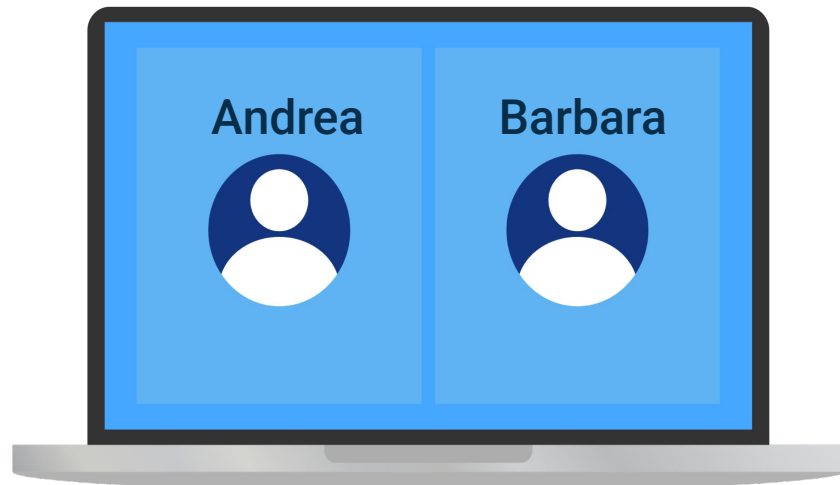
Now we'll demonstrate applying similar concepts in **Windows**.

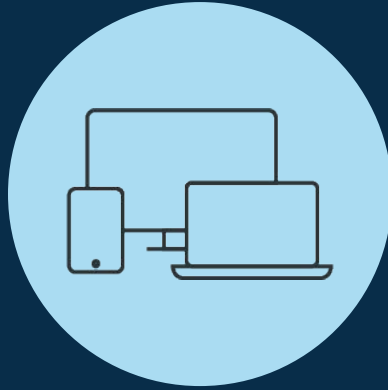


Demonstration Scenario

As we saw at the end of the previous activity, there wasn't a minimum password length set. Every organization should have a minimum password length policy.

- We'll use the `net user` utility to create Windows user accounts for two new users:
 - Senior Developer, **Andrea**
 - Sales Representative, **Barbara**





Instructor Demonstration

Adding Users and Setting Password Policies



Activity:

Users, Groups, and Password Policies

In the last activity, your CIO had you use the `net` command-line tool to find various types of information, such as the local groups on the machine and the existing password policy, with accounts.

- Your CIO noticed that the previous password policies were not very strict, and wants to enforce new password policies on this Windows workstation for the next two new users: Senior Developer, **Andrew**, and Sales Representative, **Bob**.

Suggested Time:
10 Minutes

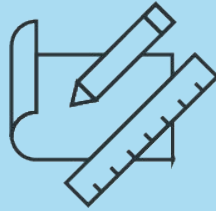


Time's up!
Let's review



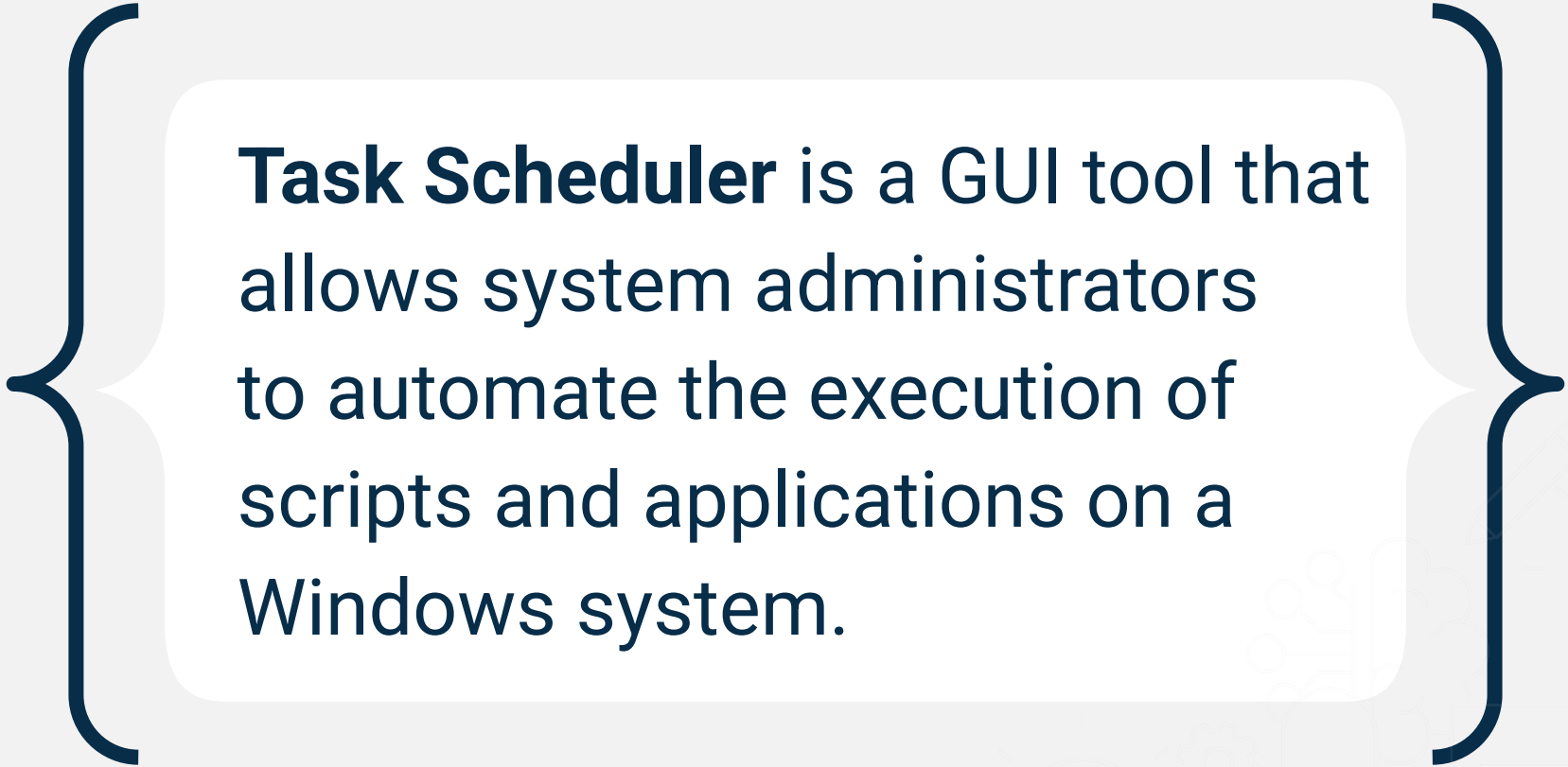
Questions?





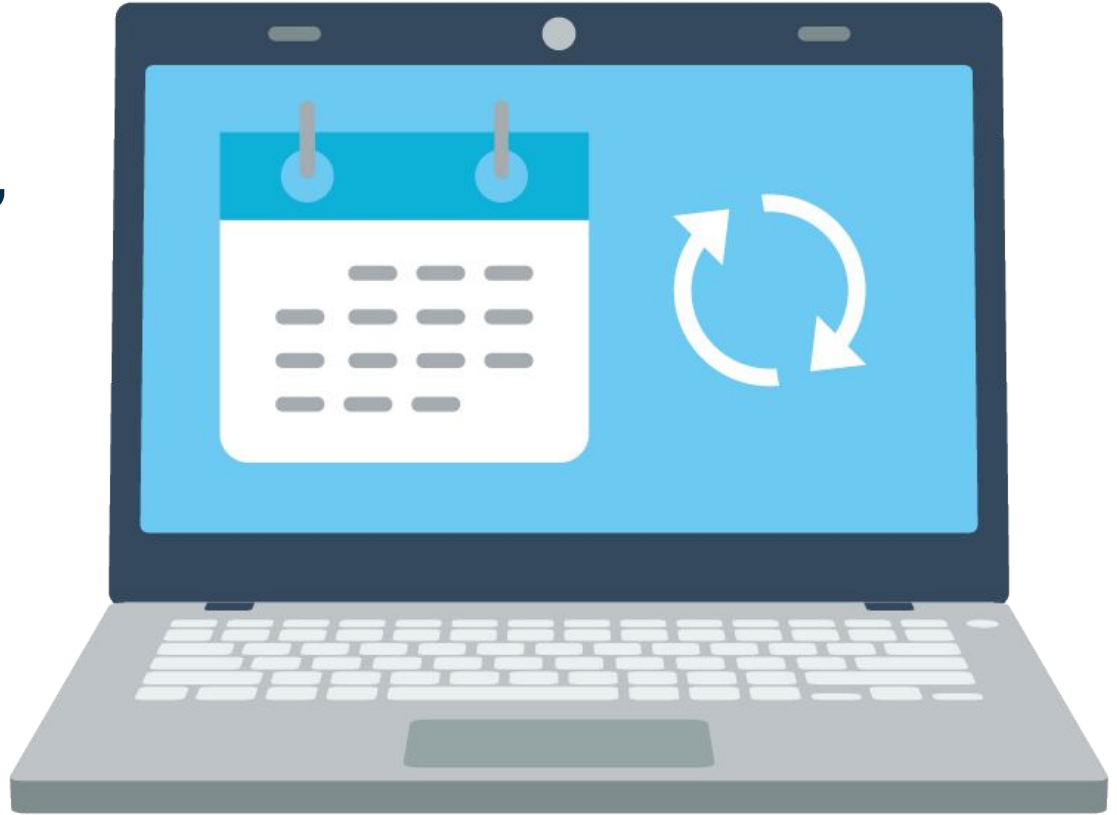
Task Scheduling

(Optional/Time Permitting)



Task Scheduler is a GUI tool that allows system administrators to automate the execution of scripts and applications on a Windows system.

Similar to **cronjobs**, tasks can be set to execute at specific times, or a certain amount of time after a user logs in.





Task Scheduling

Automating security and system administration



**Checking for
updates for
endpoint security
software**



**Sending
important logs
to systems such
as SIEM**



**Scheduling
system
maintenance
scripts**

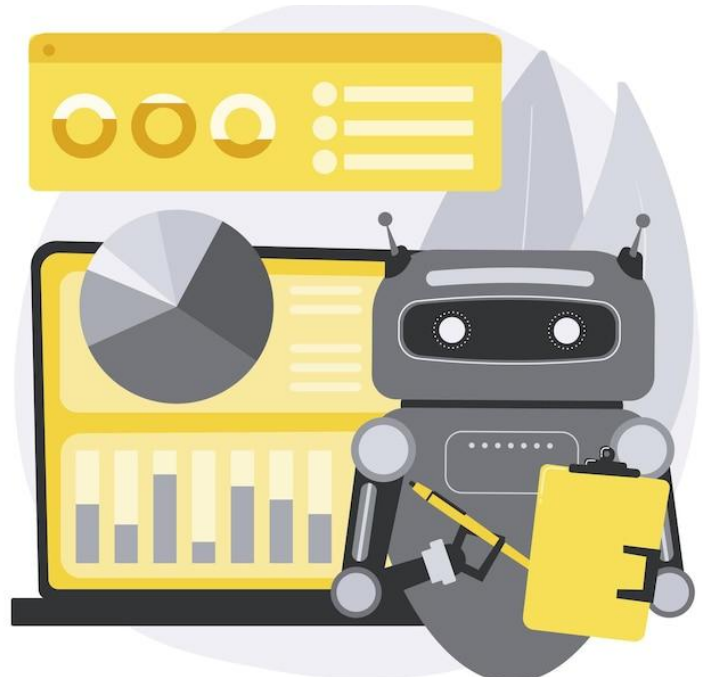


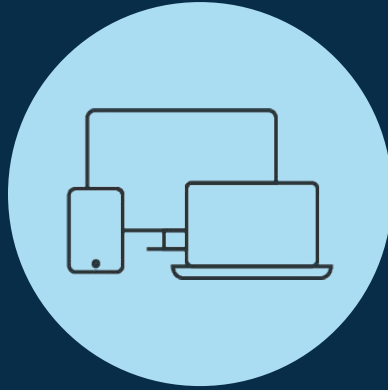
Demonstration Scenario

Your CIO wants to schedule reports to be created on a daily basis. This system isn't remotely managed at the moment, and with a new developer coming in, huge changes might be made to the machine's configuration.

This will help reduce workstation development drift for any new users for the Windows environment, and for this specific developer position and project.

Your CIO has asked you to use the GUI tool **Task Scheduler** to create a task that will run each day.





Instructor Demonstration

Task Scheduling



Time's up!
Let's review



Questions?





Next class

PowerShell Scripting



Thank you